



# SIMATS Online Programming Lab

## DS PROGRAMMING



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### QUESTIONS

10 / 10 completed

Binary Search

### BINARY SEARCH

"Write a C program implementing binary search on a sorted array of 15 integers. The program must: (1) accept an unsorted array and sort it first, (2) perform binary search, (3) display search

iterations with mid-point calculations, (4) show final result with comparison count, and (5)

verify the array remains sorted post-search.

Sample Input/Output:

Enter 15 numbers: 50 30 10 90 40 70 20 80 60 5 45 35 25 55 15

Sorted array: 5 10 15 20 25 30 35 40 45 50 55 60 70 80 90

Enter target: 45

Binary Search Iterations:

low=0, high=14, mid=7, arr[7]=40

low=8, high=14, mid=11, arr[11]=55

low=8, high=10, mid=9, arr[9]=50

low=8, high=8, mid=8, arr[8]=45

Found at index 8 (1-based: 9)

Comparisons: 4

## PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2
3 int main() {
4     int arr[15];
5     int n = 15;
6     int target;
7
8
9     printf("Enter 15 numbers: ");
10    for (int i = 0; i < n; i++) {
11        scanf("%d", &arr[i]);
12    }
13
14
15    for (int i = 0; i < n - 1; i++) {
16        for (int j = 0; j < n - 1 - i; j++) {
17            if (arr[j] > arr[j + 1]) {
18                int temp = arr[j];
19                arr[j] = arr[j + 1];
20                arr[j + 1] = temp;
21    }
```

## OUTPUT

```
Enter 15 numbers: Sorted array: 5 10 15 20 25 30 35 40 45
50 55 60 70 80 90
Enter target: Binary Search Iterations:
low=0, high=14, mid=7, arr[7]=40
low=0, high=6, mid=3, arr[3]=20
low=0, high=2, mid=1, arr[1]=10
low=0, high=0, mid=0, arr[0]=5
Not found
Comparisons: 4
Array remains sorted: Yes
```

## INPUT

50 30 10 90 40 70 20 80 60 5 45 35 25 55 15

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